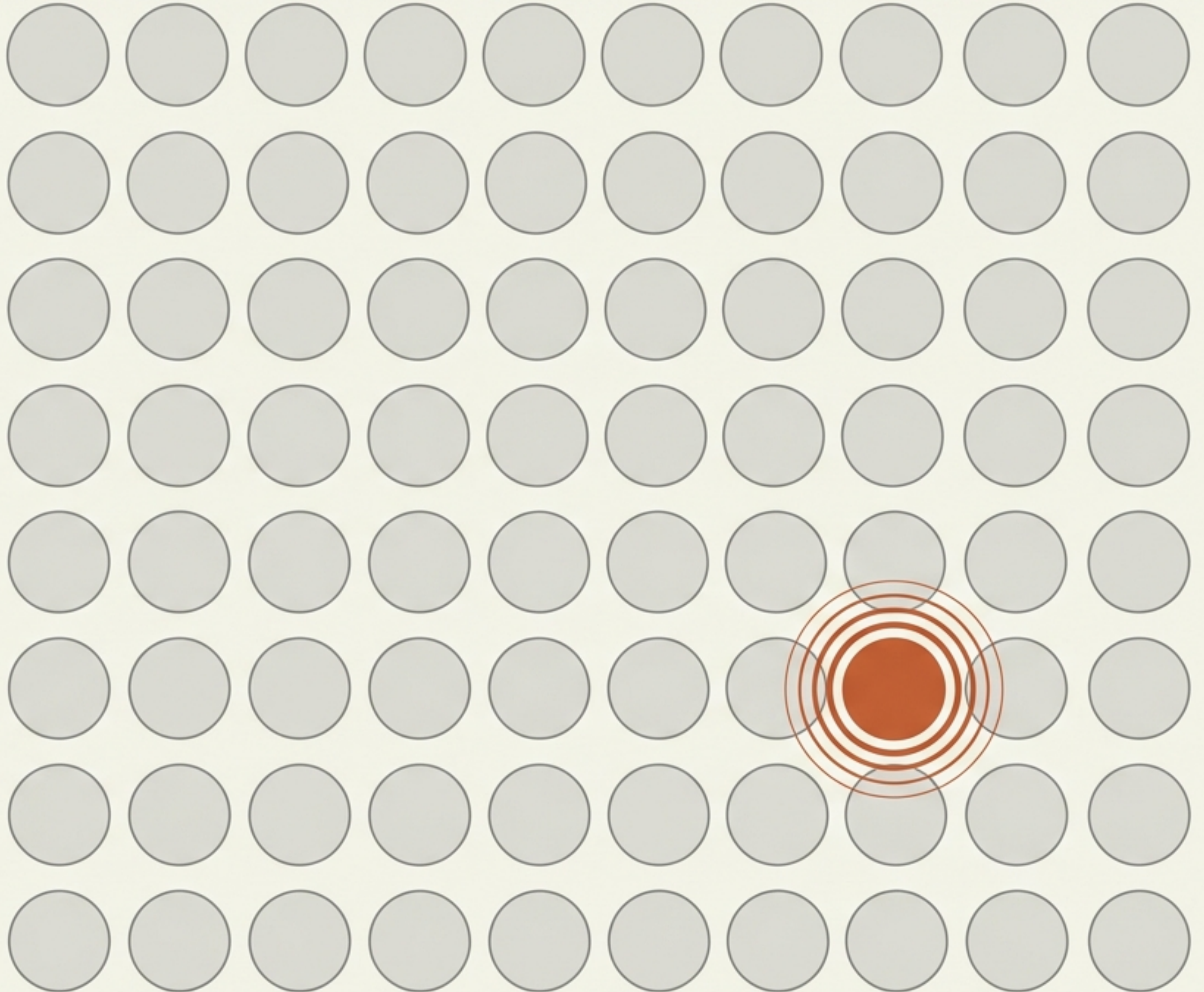


Proving Identity in the Age of Agents

Author: Dinis Cruz

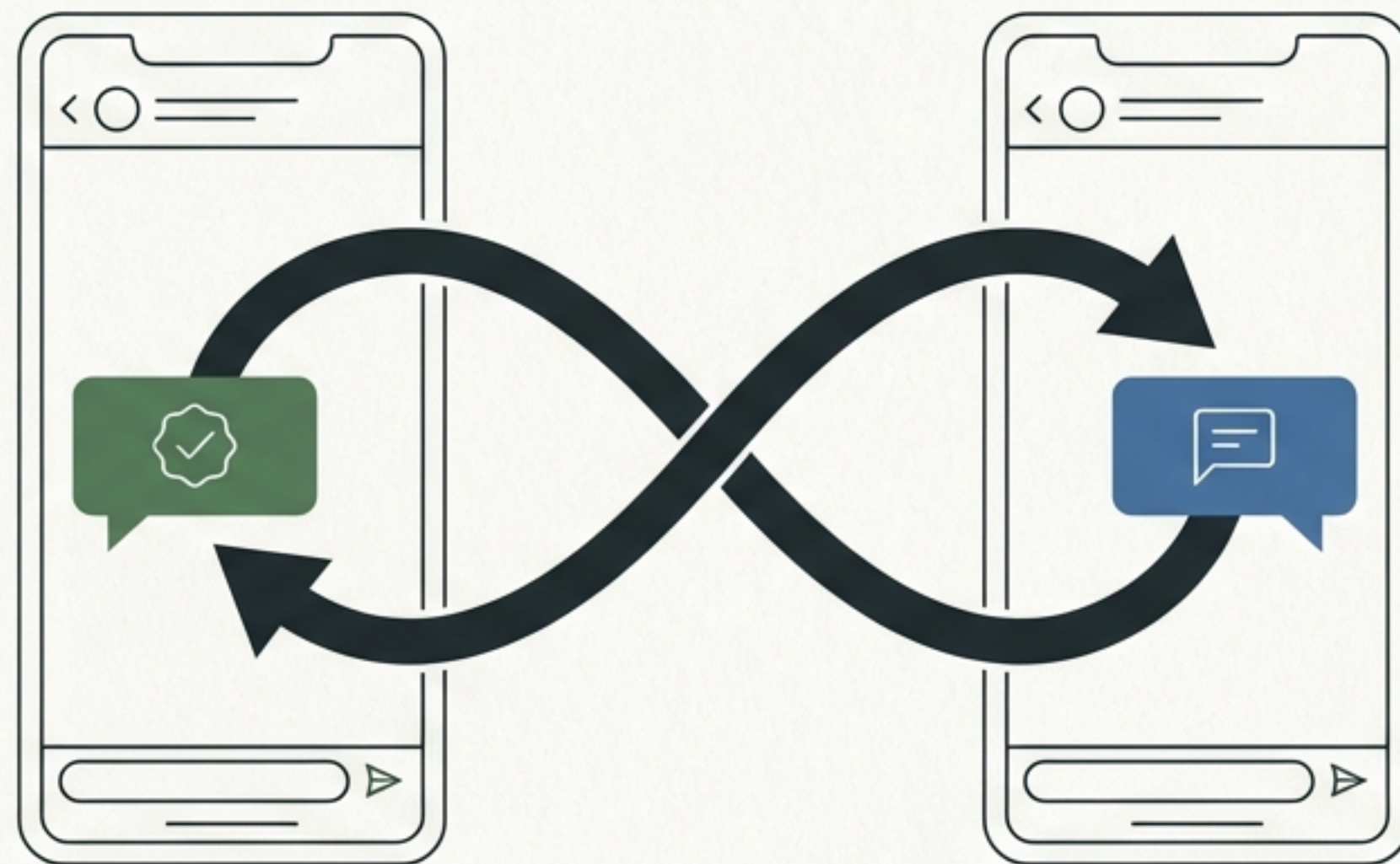
A blueprint for mathematical verification, the cryptographic web of trust, and securing machine-to-machine economies.



The modern security instinct exposes a fundamental gap

- **The Catalyst:** Reaching out to security professionals to offer product trial credits.
- **The Response:** Hey Dinis — just double checking its legit.
- **The Action:** A WhatsApp message prompts a LinkedIn verification. A LinkedIn message prompts a WhatsApp verification.

The Unsolved Problem: This is exactly what security professionals should do, but it reveals a massive systemic flaw. How do I mathematically prove I am who I say I am?

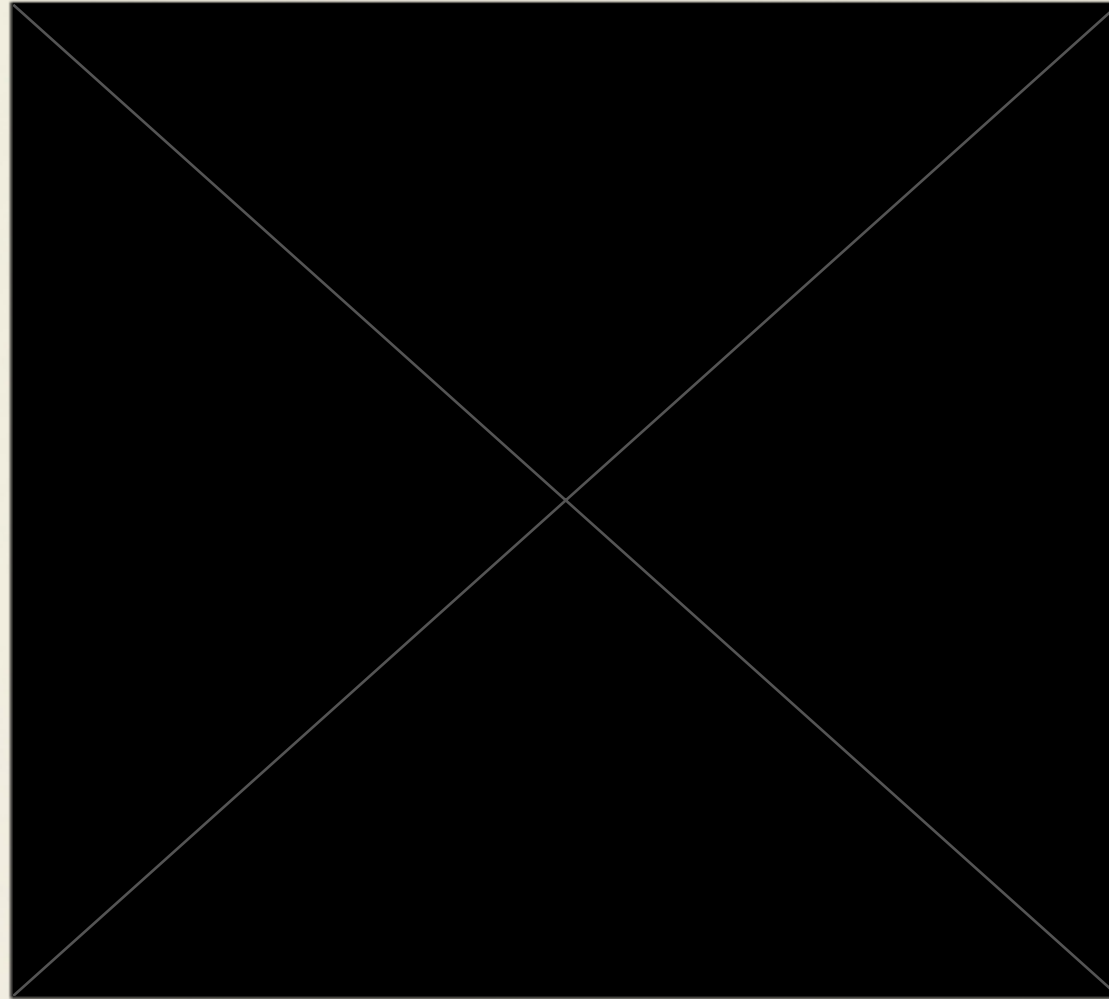


We currently rely on social signals, not cryptographic proof



LinkedIn Profiles

No real verification process.
Anyone can create a profile
claiming to be anyone.

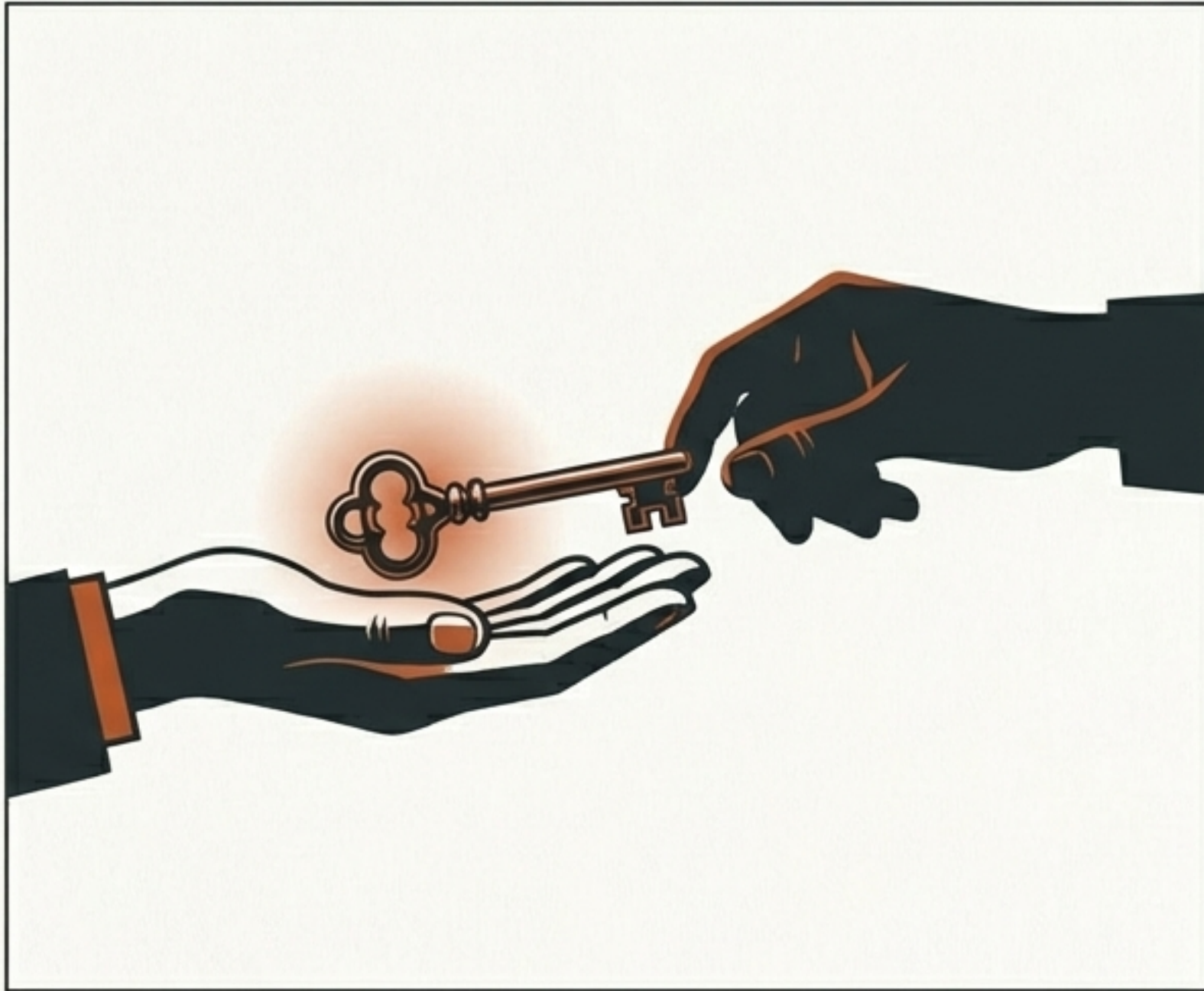


Phone Numbers

Calling a number only proves
the device is real, not the identity
of the person holding it.

**The Reality: The strongest evidence of existence is consistent activity
across platforms over years. years. These are merely social signals.**

The Identity Paradox makes users uniquely vulnerable



- Our identity systems are fundamentally backwards. Proving who you are currently requires sharing secrets.
- Publishing a phone number gives scammers a target to spoof.
- Sharing a National Insurance number hands impersonators the exact key they need.
- Distributing an email address makes you a prime target for phishing.

**Proving my identity
should make my identity
harder to steal, not easier.**

The necessary shift to public key cryptography

Fragile Present



- Relies on shared secrets.
- Identity verification exposes vulnerabilities.
- Centralised platforms hold the authority.

Cryptographic Future

Signature = $f(\text{Key}_{\text{priv}}, \text{Message})$ | Verify = $g(\text{Key}_{\text{pub}}, \text{Message}, \text{Signature})$

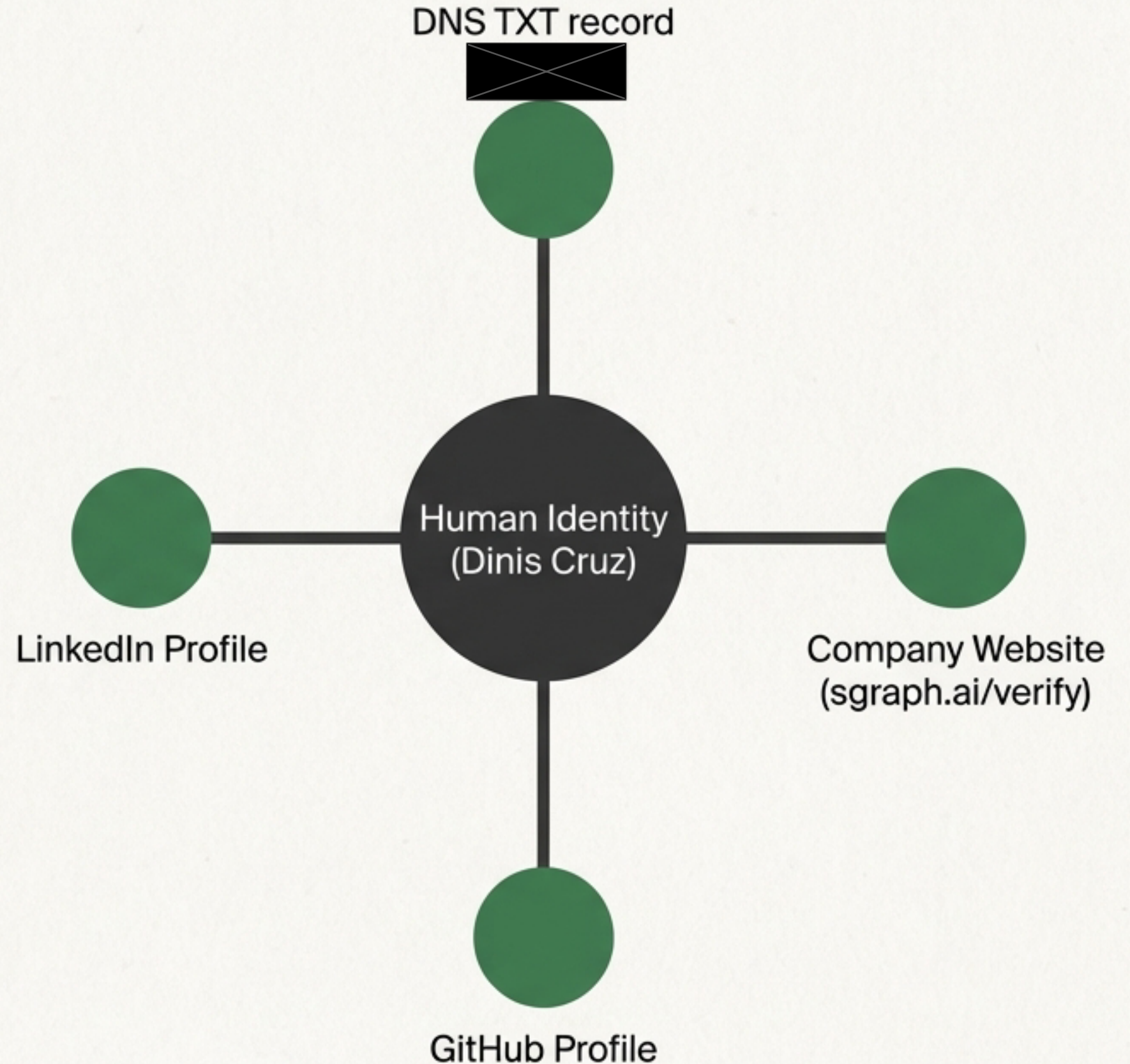


- Relies on mathematics.
- A public key is published in multiple trusted locations.

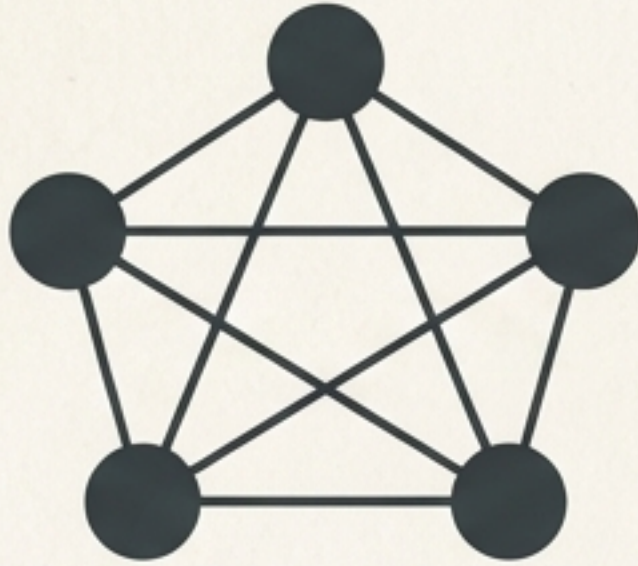
The Mechanism: Anyone can verify a signed message without the sender revealing a single secret.
Mathematics proves the identity without exposing it.

Building an interlocking web of trust

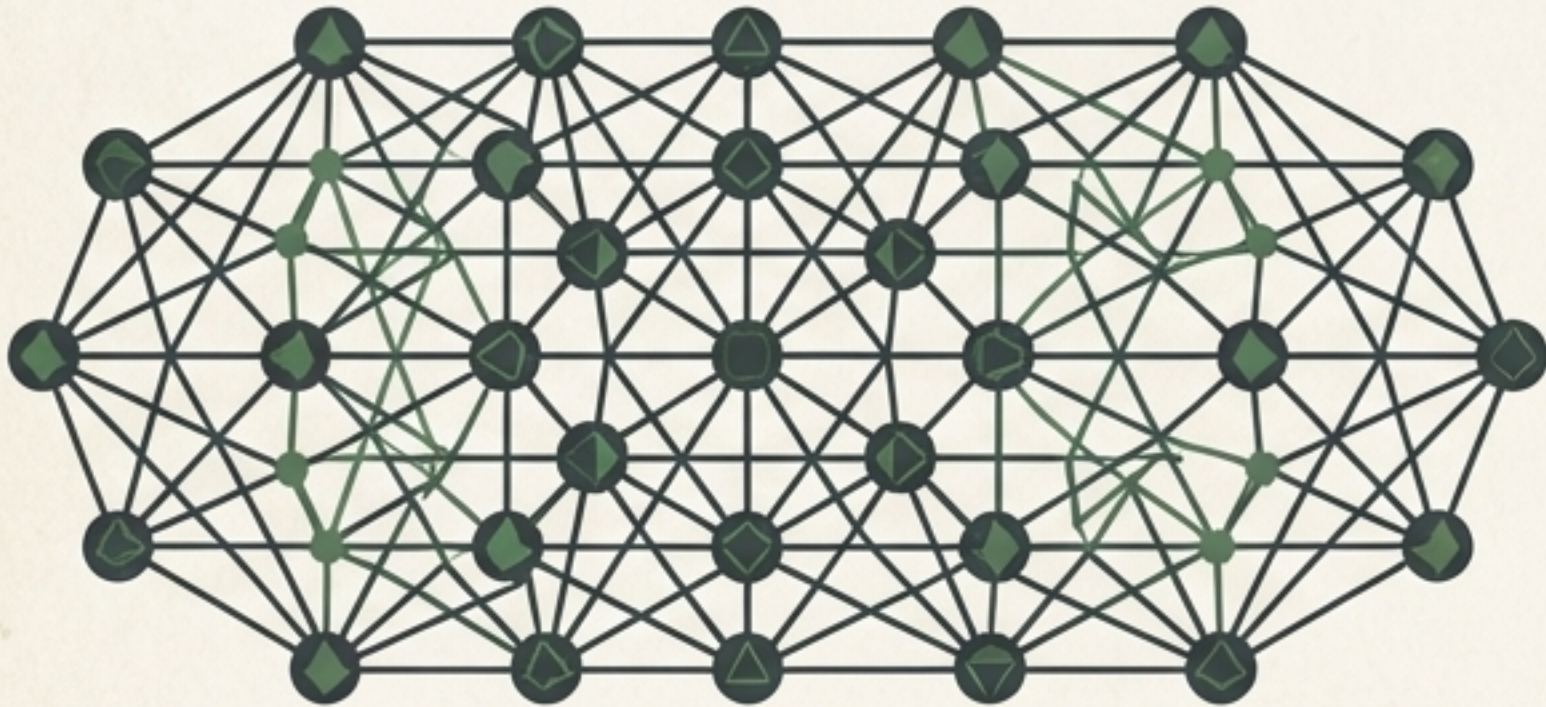
- The solution is not a single verification authority.
- Trust scales when a public key is published across multiple independent vectors.
- The Defence: To impersonate this identity, an attacker must simultaneously compromise the DNS provider, web hosting, GitHub account, and LinkedIn account. Every additional verification point mathematically strengthens the web.



Scaling the informal security community



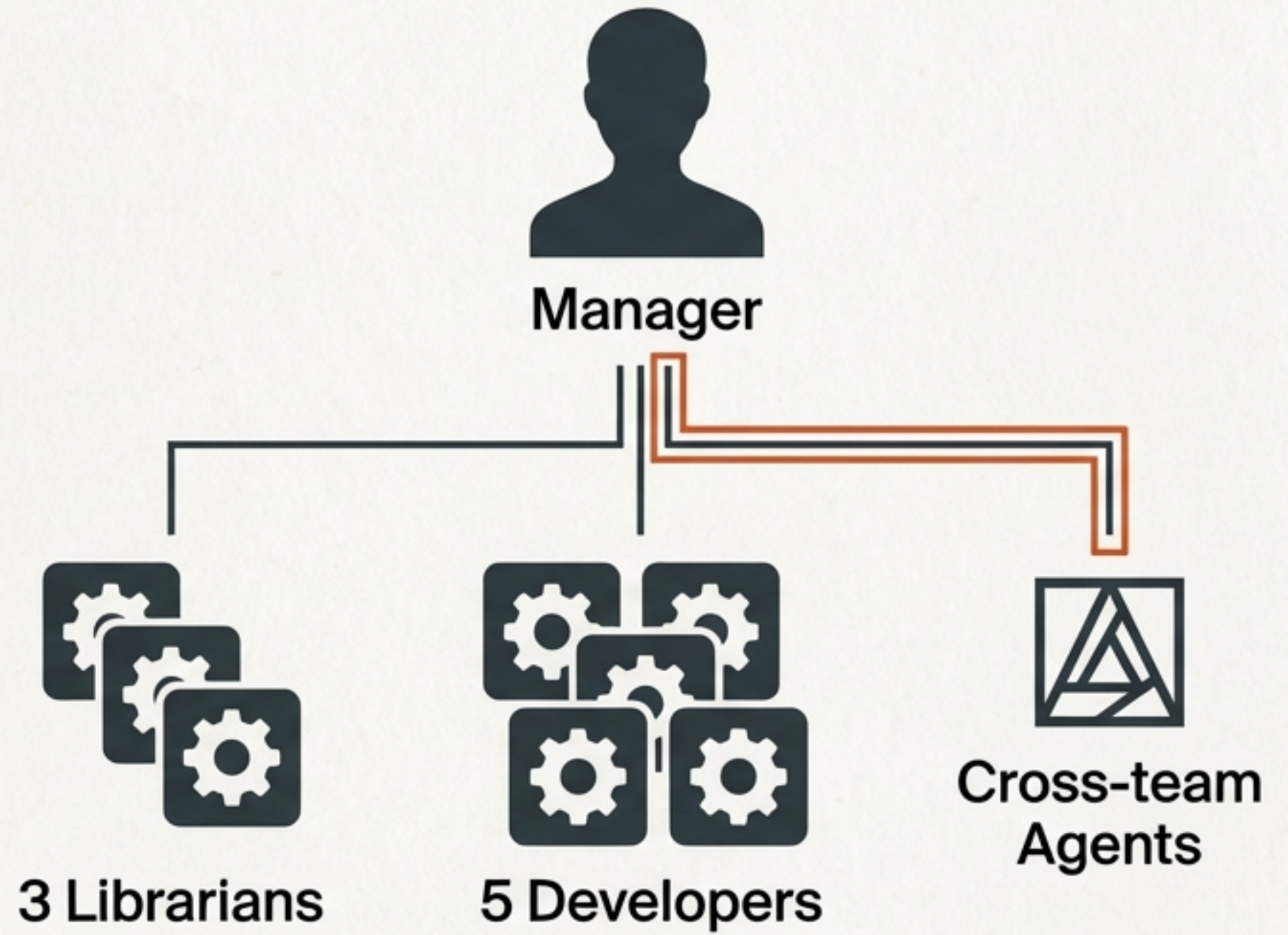
- Security communities already operate on a web of trust built through consistent participation in shared channels.
- When suspicious messages arrive, the group verifies:
Yes, that's really him or
His account might be compromised.



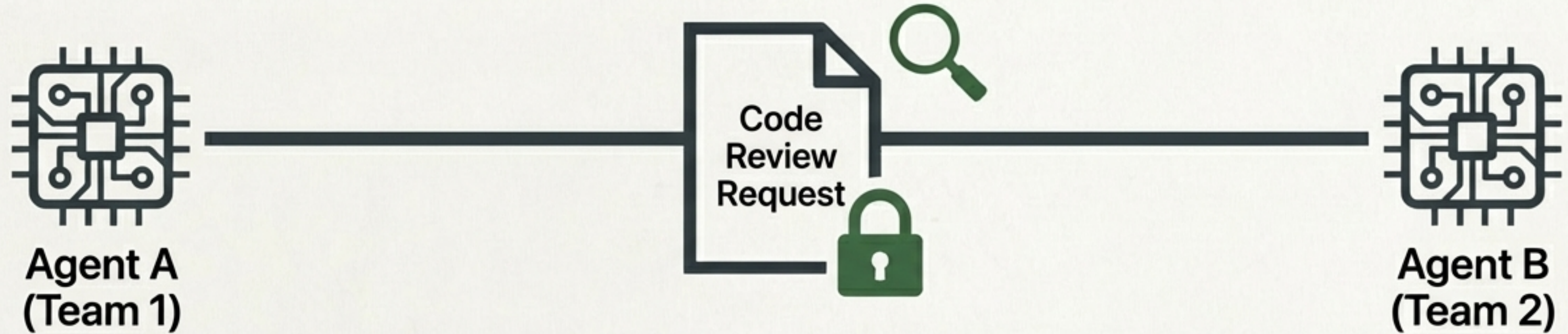
- The Limitation: This existing web is informal, invisible, and fails to scale beyond personal networks.
- We must make it formal, visible, and mathematically scalable.

Autonomous agents demand verifiable identities

- This is no longer just a human problem.
- **The Reality:** The author runs a team of distinct AI agents, each with specific roles, responsibilities, and system access.
- **The Incident:** Accidentally delegating a task to the wrong Claude window—literally talking to the wrong agent.
- If agents are to act on our behalf, they absolutely require their own verifiable identities.



Machine-to-machine trust requires mathematical proof



- When Agent A asks Agent B for a code review, Agent B must instantly verify authorisation.
- Agents cannot check on LinkedIn.
- **The Mechanism:** Each agent is issued a key pair. All inter-agent communication is cryptographically signed.

**Verification is
mathematical,
not social.**

Two gravitational forces make this infrastructure inevitable



Force 1: Agent Proliferation. As delegation to AI scales, agents must communicate, authenticate, and transact. An agent buying a service requires a verifiable identity layer.

Force 2: Deepfake Escalation. AI makes generating realistic fake messages, voices, and video trivially easy.

The Only Defence: Visual and audio proof are obsolete. Does this message have a valid signature from a key I trust? is the only question that matters.

Immediate steps to build human verification

1

Publish Everywhere

Distributing public keys across DNS TXT records, personal websites, and GitHub profiles.

2

Centralise the Evidence

Creating a dedicated **verification** hub (sgraph.ai) where anyone can mathematically audit the identity claims.

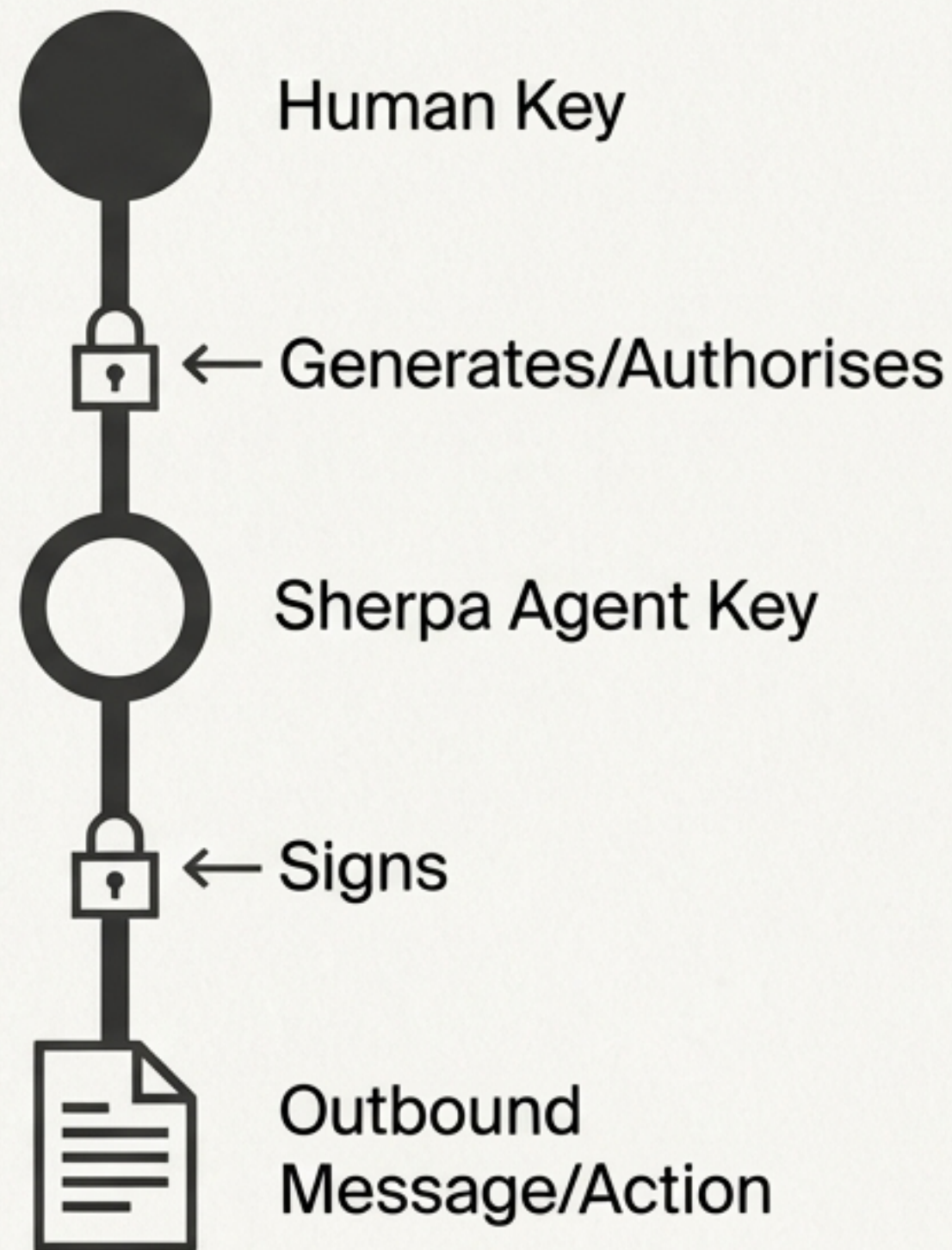
3

Sign Everything

Committing to cryptographically signing outbound messages wherever technically possible.

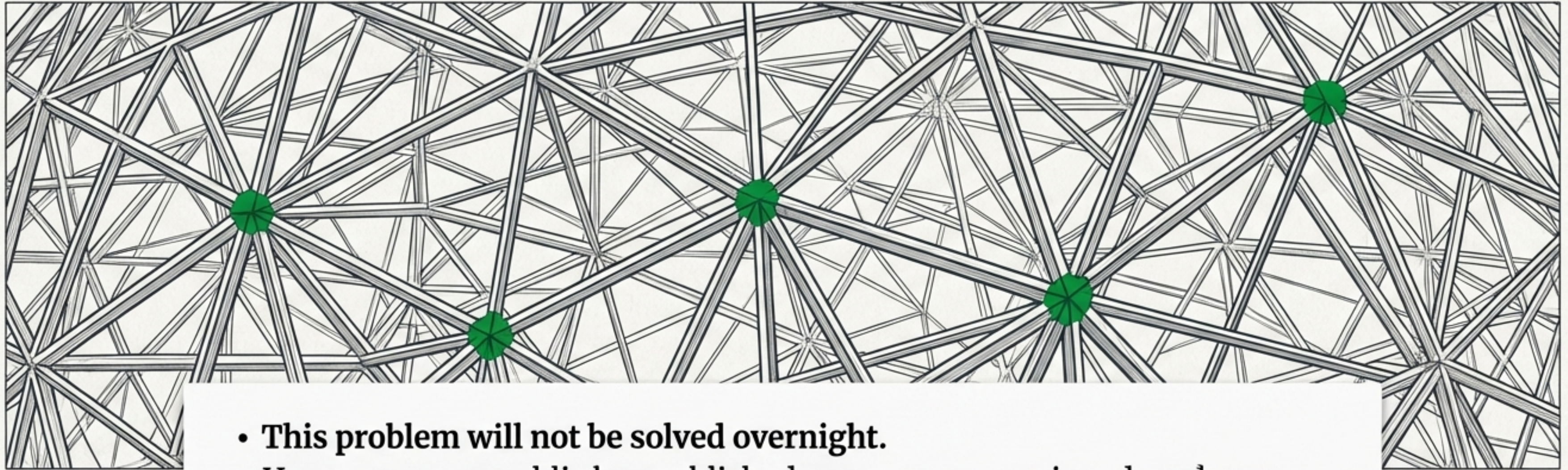


Establishing the agent chain of trust



- Every agent deployed receives its own unique key pair.
- All communication between agents is strictly signed.
- When an agent (e.g., a Sherpa) sends a message on a human's behalf, the recipient traces a pristine chain of trust.
- The recipient verifies the message was explicitly authorised by the human owner.

Weaving an unbreakable web



- **This problem will not be solved overnight.**
- However, every public key published, every message signed, and every agent identity created adds a new strand to the web.
- **Once woven, this cryptographic web is infinitely harder to break than any single password, verification step, or platform guarantee.**